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A TYPOLOGY OF FREEMIUM BUSINESS MODELS FOR MOBILE APPLICATIONS

Research

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Abstract

Freemium represents an essential business model in the accelerating mobile applications industry. While Freemium models contribute more than 90% of the revenue generated in Google Play and Apple's app store, extant research lacks to date a systematic typology. Therefore, this paper aims at developing a classification scheme for Freemium business models. Using theoretical sampling, we identified 40 apps from which we derived an empirically grounded typology comprising nine distinct Freemium types. We differentiated each type by two key themes, which are its premium revenue stream and the relationship between the free and premium offering. The developed typology contributes to a better understanding of the differentiating criteria of Freemium business models and enables risk-return considerations in the decision-making of app publishers. In addition, the typology provides a generalizable framework for different kinds of Freemium offerings. Finally, we developed three testable propositions from the typology which relate to the monetary advantage of certain Freemium types. These propositions give strategic advice on how to position existing Freemium offerings in quasi-monopolistic markets.

Keywords: Freemium business models, Digital business models, Mobile applications, Typology

1 Introduction

The importance of Freemium business models to monetize mobile applications (thereafter: mobile apps) has increased rapidly in recent years. In the mobile app industry, Freemium business models are the dominant monetization option, contributing more than 90% of revenue generated in the two leading platforms: Apple's App store and the Google Play store (App Annie, 2014; Distimo, 2013). For example, the revenue share of Freemium in Apple's U.S. App store increased from 46% to 81% in less than two years (Distimo, 2014a). Some research has been conducted on Freemium business models (Wagner et al., 2014). For example, Dörr et al. (2010, 2013) focus on the Freemium specifics of Music as a Service. Gassmann et al. (2014) and Osterwalder & Pigneur (2010) provide a conceptual analysis of the Freemium business model and its components. Oestreicher-Singer & Zalmanson (2013) and Wagner et al. (2014) examine determinants of premium conversion. Wagner et al. (2013) analyse the advertising impact of the free on the premium offering. Liu et al. (2012) assess the impact of the Freemium model on revenue. However, extant studies remain fragmented and restrict findings often to a narrow research scope. Moreover, Freemium research broadly lacks generalizable theories.

Our review of the Freemium literature (see section 3) reveals that there is no academically developed typology which derives from empirical observations. The absence of such a typology constitutes a research gap. Therefore, the purpose of this research paper is to develop a Freemium typology based on empirical findings. We focus on mobile apps as context of the study. Freemium models are crucial for the monetization of the global app industry and may therefore serve as a reasonable approximation for analysing the conceptual links of the Freemium business model. Moreover, mobile apps are available on single platforms and therefore permit easier data collection and analysis than other Freemium offerings. Consequently, we structure our research in two subsequent phases:

1) Exploration: What are significant dimensions of the Freemium business model? Which essential characteristics define those dimensions?
2) Abstraction: What are predominant themes building the conceptual foundation of Freemium business models?

This process provides the basis to answer the main research question of this paper:

> *How do central themes integrate into a systematic typology of Freemium business models for mobile applications?*

Following a broad literature review, we use theoretical sampling to analyse 40 apps which lead to the development of a novel Freemium typology. The remainder of the research paper is structured as follows: Section 2 explains the industry context and the basic business model choices for mobile app publishers. Subsequently, the relevant Freemium literature is reviewed in section 3. Section 4 outlines the research method as well as data collection and data analysis. The findings and the final typology are presented in section 5. Section 6 evaluates and conceptualizes the categorization of Freemium types and proposes future research areas. Finally, section 7 summarizes the results of the paper.

2 Mobile Applications: Industry context & monetization options

Mobile apps can be defined as application software, which users download and use on handheld devices like smartphones and tablets (Liu et al., 2012). In 2015, global mobile app sales amount to approximately 40.5 billion USD, excluding 9 billion USD in sales from mobile subscriptions and 34 billion USD in sales from mobile advertisements (VisionMobile, 2015). By 2017, the consumer Apps' market size is estimated to exceed 74 billion USD (Digi-Capital, 2015). The two central mobile app platforms are the Android-based Google Play Store and the iOS-based Apple App store. Together, these two platforms account for 96.3% of the global market, while Google Play alone contributes 81.5% (IDC, 2015). In terms of app downloads, Google Play outperforms Apple's App store by almost 60%, whereas Apple globally generates 70% more revenue with its App store than Google Play

does (App Annie, 2014). In Germany, however, Google Play's revenue has been outgrowing the Apple App store and now even leads the German market in terms of total revenue (App Annie, 2015a). Globally, games are by far the most significant app category in both Google Play and the Apple store (Distimo, 2014b; 2013), aggregating the most downloads and the highest sales levels. The top three app categories, in terms of downloads, are completed by 'Communication' and 'Tools' for Google Play, and by 'Entertainment' and 'Photo & Video' for the Apple App store (Distimo, 2014b).

Based on a meta-analysis of seven field-relevant sources (Google Play, 2015a; App Annie, 2014; Pierce and Wooldridge, 2014; Distimo and MEF, 2013; Distimo, 2013 & 2014a; Adobe Systems Inc., 2012), we find that mobile app publishers can fundamentally choose one of the four possible monetization options (see figure 1). First, the customer can simply purchase the app and continuously consume it without any additional charge (One-time app sales). Second, the customer initially pays for the download of the app and has the option to upgrade ("Twinned" premium). Third, the user downloads and consumes the app entirely at no cost without any time or feature constraints (Free to the user). Fourth, download and consumption can be completely free, but the user has the option to extend, among others, usage time after free trial, functionality and user experience (Freemium; adopted from App Annie, 2015b).

Downloads		No	Yes
	Paid	(1) One-time app sales	(2) "Twinned" premium
	Free	(3) Free to the user	(4) Freemium
		In-app purchases	

Figure 1. Overview of Monetization options (Source: Own illustration based on meta-analysis)

While One-time app sales and "Twinned" premium are easy to grasp, free-based business models offer multiple variations that have to be clarified. "Free to the user" does not necessarily imply that no monetization takes place. The user base is often indirectly monetized via advertisements financed by third parties, direct cross-subsidies of free apps which are subsidized by other paid offerings (Anderson, 2009), or by selling user data (Gassmann et al. 2014). Only in a so-called "gift economy" model (Anderson, 2009), apps operate without any monetary charges (e.g. Wikipedia). Moreover, free-based models create hybrid models. For example, Music streaming service providers (e.g. Spotify) often apply a hybrid model which monetizes the user base by combining Freemium and ad-supported third-party models (Dörr et al., 2010). Hybrid models are likely to exist across all monetization options.

In addition, free versions might supplement paid versions. This hybrid model is defined here as "versioning". Customers self-segment by choosing between the free basic version and the more sophisticated, paid version based on their individual willingness to pay (Shapiro & Varian, 1998a). In case of mobile apps, the publisher can additionally opt to allow for entirely free of charge app consumption or to integrate in-app purchases. The app "InstaWeather", for example, is offered both as a free basic version with in-app purchases (Google Play, 2015b) as well as a paid version ("Pro") that also comprises in-app purchases (2015c). Therefore, versioning encompasses both up-selling versions from "Free to the user" to "One-time app sales" and up-selling from "Freemium" to "Twinned premium". The latter type establishes versioning as the broader term compared to Freemium. While versioning spans across all basic monetization options and implies the creation of two separate apps, we define Freemium apps here as a separate monetization option to be distributed as a single offering.

3 Freemium business models

In this section, we present the main literature. "Freemium" is a composition of the terms "free" and "premium" which implies the coexistence of free and paid elements (Gassmann et al., 2014; Wagner et al., 2014). Thus, Freemium generally combines a basic product or service free of charge with more sophisticated product components or advanced services which are charged (Seufert, 2014). We identified diverse conceptualisations for the term Freemium. For example, Bekkelund (2011) provides a collection of different meanings implied by the term "Freemium", including Freemium as a

marketing instrument, a distribution model and as a revenue model. By analysing the conceptual linkages, it can be shown that Freemium is a business model. Osterwalder & Pigneur define a comprehensive business model as "the rationale of how an organization creates, delivers and captures value" (2010, p. 18). Since Freemium differentiates between free and premium offering in its value proposition (Osterwalder & Pigneur, 2010), it is likely that Freemium goes beyond a pure revenue model. In their research on fifty-five distinct business models, Gassmann et al. (2014) identified value proposition and profit mechanism as the hallmark business model components of Freemium. While value proposition is understood as the firm's offerings to satisfy customer needs, the profit mechanism addresses the commercial feasibility of the business model and comprises its cost structure and revenue model (Gassmann et al. 2014). Hence, Gassmann et al. (2014) established that Freemium combines value creation (free and premium offering) and value capture (profit mechanism). We consequently define Freemium as a business model, as it comprises two constitutive components with value creation and value capture.

Moreover, the distinction between free users and paying customers is important to explain the Freemium business model (Osterwalder & Pigneur, 2010). Free users receive the basic offering without a charge. In return, free users might later convert to paying customers and engage in word-of-mouth marketing (IVP, 2012). Interestingly, Gupta and Mela (2008) found out that free users tend to reach their maximal monetary value in a company's early stages because they are essential in the formation of network effects. Nevertheless, free users face limitation mechanisms that serve to distinguish the premium offering (based on App Annie, 2015b; Pujol, 2010; Anderson, 2009). The free offering can be restricted with regard to its functionality, so the customer pays for advanced features, additional content or services, upgrades, speed-ups, and virtual currency/items. Moreover, customers may be charged after completing a free trial period. The customer can also pay to eliminate or augment any barrier related to app usage (e.g. time, storage space). In addition, customers may be charged if they desire to improve their user experience by removing advertisements. Finally, any combination among these four limitation mechanisms is possible (see table 1).

Limitation mechanism	**Value proposition**
Functionality / Feature-limited	The premium offering provides virtual currency, "virtual items, speed-ups, content, add-ons, upgrades, services or capabilities" superior to the free offering.
Free Trial / Time-limited	The free offering already permits access to full functionality, but for a restricted time period. The user has to convert to premium to continue using the (premium) offering after the termination of the free trial period.
Usage	The free offering has "limited usage, bandwidth, hours or storage space". The premium offering reduces or eliminates those barriers.
User Experience	While the free offering is financed by advertisements, the premium offering allows for an elimination of these advertisements.
Combinations	Any combination of the four mechanisms above.

Table 1. Choices of Limitation mechanisms (App Annie (2015b), Pujol (2010), Anderson (2009))

In contrast, paying customers are granted premium access rights to superior app elements in exchange for money (adopted from Osterwalder & Pigneur, 2010). Often, a relatively small group of paying customers subsidizes a large free user base (Osterwalder & Pigneur, 2010; Anderson, 2009). The premium conversion rate is usually a single percentage digit (Gassmann et al., 2014). In practice, a benchmark rate of 5% of all users who eventually convert to paying customers is frequently used (Seufert, 2014). The price of zero for the free offering maximizes the group of potential users, whereas the premium offering generates revenue (Seufert, 2014). Freemium customers are often characterized as the "professional" (Osterwalder & Pigneur, 2010), "most engaged", "most enthusiastic", "passionate" and "dedicated" user group (Seufert, 2014). A central question is why free users convert to paying customers, while the basic offering is provided at no cost (Wagner et al., 2014). In an online community, for example, the most active users are significantly more likely to convert to premium customers (Oestreicher-Singer & Zalmanson, 2013). Furthermore, Wagner et al. (2014) discovered that the likelihood of premium conversion for music streaming services is higher if the free offering

corresponds closely with the premium offering. The Freemium business model provides remarkable benefits. Prospects of user acquisition are significantly enhanced because Freemium implies no initial monetary barrier to access the offering (Seufert, 2014). Both users with scarce or no disposable income and users who are not convinced of the price-value-ratio in a paid download scenario are more likely to be acquired in the Freemium scenario. Moreover, if the free user base actively engages in word-of-mouth marketing, the diffusion of Freemium offerings in the market can be considerably accelerated without charge (Jiang & Sarkar, 2009). Freemium also enables a risk-free demonstration of the free offering (Jiang, 2010) which is particularly advantageous for experience goods (Niculescu & Wu, 2014). Thereby, the free offering serves to attract users to the premium offering (Wagner et al., 2013). In particular, the free offering can result in both acceleration (i.e. users buy the premium earlier) and expansion effects (i.e. users buy the premium, although they had not contemplated payment at first) (Bawa & Shoemaker, 2004). A price of zero potentially results in a competitive advantage (Shapiro & Varian, 1998a), which makes it hard for industry incumbents to compete based on their initial price points (Anderson, 2009). For several app categories (e.g. online games), the maximization of the total user base also fosters the creation of network effects (Wagner et al., 2014). As the overall utility of the user base increases with every new user accessing the offering (Cheng & Tang, 2010), these "demand-side economies of scale" (Katz & Shapiro, 1985) are continuously extended. Nevertheless, there are various challenges of Freemium models which need to be addressed. Due to initially established consumption at no cost, the free offering may considerably hamper premium conversion (Sixteen Ventures, 2010). Instead of converting to paying customers, some users continue to use the free offering (Bawa & Shoemaker, 2004). The free offering cannibalizes the premium offering, so this cannibalization accounts for a loss in revenue (Cheng & Tang, 2010). Another central challenge is the conversion of the free user base to paying customers at sufficiently large numbers (Seufert, 2014) in order to achieve sales levels superior to other monetization options. The large group of free users depletes scarce resources which may be more effectively used for the premium customers (Sixteen Ventures, 2010). Particularly for mobile apps, Freemium publishers have to cope with low switching costs for free offerings and low levels of relatedness among their app offerings (Liu et al., 2012). Therefore, the challenge is to establish barriers to exit for the users in order to prevent users from switching among the various free offerings in the market (Sixteen Ventures, 2010).

4 Method & Data

4.1 Research method

We applied theory building from case studies to develop an empirically based Freemium typology (Eisenhardt, 1989). While typologies are often mistaken as pure classification systems (Rich, 1992), they actually meet essential theory criteria. These theoretical properties comprise the existence of constructs, relationships among constructs, and the possibility to falsify and disconfirm these constructs and their relationships (Doty & Glick, 1994,). Thus, typologies and theories are closely related. Explorative case research supports creative and interpretive thinking which is central for developing new theories (Eisenhardt, 1989). In addition, theory development in case research is closely tied to empirical data, which considerably increases the likelihood of obtaining an empirically valid theory (Eisenhardt, 1989). We relied on an iterative research process which offers the flexibility required for insight-driven data collection and analysis (Glaser, 1978). Iterations allow for adjustments in the research structure during the study, in case the data does not fit the initially assumed structure. Furthermore, the volume of data to be examined is greatly reduced, so enabling us to apply deep data analysis (Glaser & Strauss, 1967). Consequently, theory building from case studies offers several benefits which are critical for the generation of a novel Freemium typology.

We used theoretical sampling in this iterative, non-linear research process of simultaneous data collection, coding and analysis (Glaser & Strauss, 1967; Martin & Turner, 1986). Since theoretical sampling is exclusively guided by the emerging theory, the key decision criteria for the sequence of

the process are theoretical purpose and relevance (Glaser & Strauss, 1967). While theoretical purpose focuses on which new data best serves the defined research purpose of the sampling process, theoretical relevance determines which additional data is the most decisive to effectively develop theory. Theoretical saturation of the most important dimensions is the central decision criterion for closing the sampling process (Glaser & Strauss, 1967). As a complete descriptive coverage of data might result in data redundancies, theoretical saturation targets the conceptual level of theory development (Breckenridge & Jones, 2009). The theoretical saturation point is reached when additional data would not yield essentially new information (Hallberg, 2006). More specifically, there are three qualifications leading to theoretical saturation (Ng & Hase, 2008, based on Strauss & Corbin, 1990). Theoretical saturation is achieved when (1) adding new cases would not create any new dimension; (2) all dimensions, particularly those with the most explanatory power, are richly and densely described with sufficient coverage of their fundamental characteristics, and (3) linkages between dimensions are substantiated and adequately backed up by empirical data. We therefore iterated continuously among collecting, coding and analysing data until reaching confidence that the sample has reached its saturation and that a valid theory has emerged (Breckenridge & Jones, 2009).

Despite the iterative nature of the typology building process, academic writing conventions require sequential research documentation with separate sections for data collection and analysis (Suddaby, 2006). While our research process followed the theoretical sampling method described above, we report data collection and analysis as distinct phases.

4.2 Data collection

The Freemium business model is our main unit of analysis in this study. Hybrids of Freemium and other business models (e.g. ad-supported models) are exclusively analysed to the extent to which respective cases (e.g. Deezer) are relevant for developing the Freemium typology (e.g. Deezer: no advertisement in the premium offering). We focused our data collection and analysis on mobile apps of the Google Play Store. Neither pure-play websites nor other app platforms were consulted while developing the typology. Conducting data collection and analysis based on field observations in Google Play, we relied on directly observable data to generate the typology. As guidance for the data collection process, researchers distinguish among three different stages of theoretical sampling (Strauss & Corbin 1990; Coyne, 1997). First, open sampling attempts to maximize early learning by analysing the theoretically most relevant cases. Variational sampling seeks to identify differences across cases to saturate theoretically relevant dimensions. Ultimately, discriminate sampling completes theoretical sampling by saturating the remaining, weakly developed dimensions and validating the overall theory. We followed this sampling process and selected an elaborate starting point for data collection to produce first theoretical insights (see Breckenridge & Jones, 2009). Therefore, we relied upon open sampling to work off the list of top 10 downloaded Freemium apps in descending order (see Lee & Raghu (2014) for a similar approach). Since 7 out of the top 10 Freemium apps were games, the analysis of apps from additional app categories (e.g. Communication, Music & Audio) resulted in the highest progress regarding the exploration of dimensions and their characteristics. Consequently, we sampled the top downloaded apps from additional app categories (e.g. shopping, social networks, tools) and identified the relationship between free and premium offering as the first key theme (see section 5). However, its relationships were of significantly more diverse nature than anticipated. Since apps with a high convergence of free and premium offering outnumbered the sample collected thus far, we selected apps with a 'balanced' or 'divergent' relationship in the variational sampling phase. The subsequent analyses revealed that premium revenue streams could be identified as the second key theme (see section 5), but its various differences regarding the payment structure had to be confirmed. Thus, we next focused on apps validating the second key theme. We progressively added apps to the sample until theoretical saturation was reached in accordance with the discriminate sampling strategy (Eisenhardt, 1989; Strauss & Corbin, 1990). In sum, we sampled 40 apps from 18 different app categories to establish the Freemium typology.

4.3 Data analysis

We conducted data analysis by applying the constant comparison method (Glaser & Strauss, 1967) which systematically captures differences and similarities (Spiggle, 1994). Theory building from case studies starts with a detailed within-case analysis (here: individual apps) that is followed by broader cross-case comparisons to derive the most essential themes (Eisenhardt, 1989). Table 2 summarizes the hierarchy of the constructs used for the two research phases.

Constructs used in *exploration* phase	**Definition in this paper**	**Examples**
Dimension	Distinctive parameter that describes each Freemium offering	Revenue model
Characteristic	The specific attribute(s) of each dimension	Subscription, pay-per-use, micro-payments
Constructs used in *abstraction* phase	**Definition in this paper**	**Examples**
Theme	Overarching concept that builds the foundation of the final typology	Premium revenue streams
Population	Homogeneous group within each theme that shares certain elements	Fixed, merged, variable

Table 2. Overview of constructs used in respective research phase

Applying a broad understanding of the term "data" (Glaser, 2001), we integrated the most significant dimensions identified in the literature on Freemium business models as conceptual basis for data collection and analysis (see table 3). The "a priori specification of constructs" allows for more accurate measurement and, provided that these constructs are relevant to the final research result, improves the empirical foundation of the resulting theory (Eisenhardt, 1989). The dimensions identified in the literature review needed to be directly observable to match the chosen research method.

Key dimensions identified in literature	**Sources** (ordered by year of publication)
Free offering	Gassmann et al. (2014), Wagner et al. (2013, 2014), Seufert (2014), Osterwalder & Pigneur (2010), Jiang (2010)
Premium offering	Gassmann et al. (2014), Wagner et al. (2013, 2014), Seufert (2014), Osterwalder & Pigneur (2010)
Nature of the offering (pure Freemium vs. hybrid forms)	Wagner et al. (2014), Dörr et al. (2010)
Revenue model / monetization	Wagner et al. (2014), Seufert (2014), Bekkelund (2011), Osterwalder & Pigneur (2010)
Limitation mechanism of the free offering	App Annie (2015b), Pujol (2010), Anderson (2009)

Table 3. Key dimensions derived from literature

In order to facilitate a better understanding of dimensions and their characteristics, we exemplified our data analysis of Skype in table 4 (App Annie (2015s); Google Play (2015u); Skype (2015)).

Dimensions	**Characteristics**
Free offering	1. Voice & video calls to all peers on Skype; 2. Group chats & calls; 3. Screen-in-screen calling & screen sharing; 4. Share pictures and other files; 5. Ubiquitous chatting across devices; 6) Text, voice & video messages; 7. Skype number: a portable local phone number; 8. …
Premium offering	Calls to mobile and landline phones; Send SMS
Limitation mechanism	Functionality
Nature of offering	Pure Freemium
Revenue Model	Choice between monthly, country-dependent subscriptions and pay-per-use (call minutes or SMS that charge a prepaid deposit)

Table 4. Exemplary data analysis and coding for Skype

Based on the conducted data analysis, we coded and compared all individual cases within the sample. Strauss and Corbin (1990) describe three hierarchical coding procedures with increasing levels of abstraction (Breckenridge & Jones, 2009; Ng & Hase, 2008). Following a close examination of individual cases, open coding seeks to identify dimensions and their characteristics apparent in the data. Axial coding strives for linkages between dimensions and aggregates open codes into themes. Lastly, selective coding serves to incorporate the developed themes and their populations into a systematic typology (adopted from Darke et. al, 1998).

The data analysis process is briefly outlined below. In the exploration phase, we applied open codes to describe the free and premium offering which relate to the value proposition and describe the benefits attributed to free users and paying customers, respectively. To assess the nature of offerings, we distinguished between pure Freemium offerings and hybrid forms (e.g. ad-supported offering). Furthermore, we analysed the revenue model(s) for each app. We applied axial coding to identify the limitation mechanism connecting the dimensions free and premium offering. In the abstraction phase, themes for the final typology (see section 5) were selected based on their relevance for the subject, the adequacy for development of a typology and the fit with both data sample and literature. The diversity regarding the contextual linkage between the free and the premium offering led to the theme relationship between the free and premium offering. Based on the empirically identified characteristics of the dimension 'revenue models', we grouped similarities regarding the payment structure into three distinct populations which together form the theme premium revenue streams (see table 5).

1. Exploration phase		**2. Abstraction phase**	
Dimension	**Exemplary Characteristics**	**Populations**	**Theme**
Revenue models	Sell virtual items, pay-per-feature, pay-per use	Variable	Premium revenue streams
	Pay-per-issue or subscription, flat rate and virtual items	Merged	
	One time lump-sum payment, subscription, flat rate	Fixed	

Table 5. Derivation of premium revenue streams (both Exploration and Abstraction phases)

As already identified in the literature review, both the relationship between free and premium offering and their revenue streams are hallmark attributes of the Freemium business model (Gassmann et al., 2014; Wagner et al., 2014; Osterwalder & Pigneur, 2010). Given the data sample, these two themes provided sufficient diversity to effectively distinguish among a very diverse set of Freemium types (section 5). We then applied selective coding to differentiate populations within each theme. Our analysis has shown that each theme comprises three distinct populations which together build the two-theme Freemium typology leading to nine distinct freemium types (see figure 2).

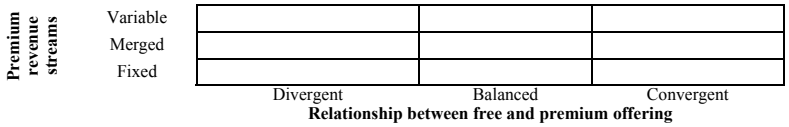

Figure 2. The two themes of the Freemium typology and their populations

5 Results

We present the main findings of our research in this section. We first define the respective theme and then describe each population. The first key theme for building the typology is the relationship between free and premium offerings. This theme describes the degree to which free and premium offering differ from or correspond with each other. Naturally, the level of convergence or divergence, respectively, is continuous and lacks clear-cut boundaries. For each app, we compared the complete offering (free and premium offering taken together) with the free offering. Thus, the location of each app on this continuum is based on a case-specific, relative assessment. To construct distinguishable populations, to form the typology, we abstracted this continuum into three populations (see table 6).

Differentiators (High divergence)	**Compromisers ('Balanced' relationship)**	**Conformists (High convergence)**
• Free and premium offering are clearly separable • Distinctive premium offering that offers significant value • (c.p.) Low monetization risk (majority of offering is paid)	• Tendency to "balance" gratis and paid offering • Clearer premium differentiation than conformists • Free offering takes up a larger portion than differentiators • (c.p.) Medium monetization risk	• Free and premium offering are closely related • Few premium benefits provide differentiation • (c.p.) High monetization risk (majority of offering is gratis)

Table 6. Explanation of populations for theme "relationship between free and premium offering"

The differentiator population shows a high level of divergence between the free and premium offering, so both offerings are rather detached from one another. Consequently, premium is clearly distinguishable from the free offering and provides significant enhancements which can optionally be purchased on top of the free offering. For example, the language learning app Babbel grants free access to one lesson for each section (App Annie, 2015h & Google Play, 2015h). The premium offering, in turn, grants complete access to all lessons for a broad variety of learning sections (e.g. Spanish has up to 22 lessons per section). Accordingly, if you want to effectively learn a new language using Babbel, you have to convert to a paying customer. This example illustrates that the differentiator population has a distinctive premium offering and provides advanced elements that add considerable value to the free offering. Since differentiators tend to be restrictive in giving away their free offering, the monetization risk is comparably low (c.p.).

In contrast, the conformist population exhibits a high level of convergence of free and premium offering, so both offerings are closely related and only some specific premium benefits enable a differentiation of the premium offering. For example, the game Candy Crush Saga only offers a virtual currency and several virtual items as premium offering (App Annie, 2015ac & Google Play, 2015ae). All game features are already included in the free offering, so the game can be played completely without paying for the premium offering. The difference to the free offering is a time-wise acceleration of the gaming progress if the premium offering is bought. Thus, the conformist population provides the vast majority of its app offering without charge and monetizes only a narrowly defined premium offering. As most of the offering is given away for free, a high level of convergence carries a higher monetization risk compared to differentiators (c.p.).

Between the two extreme populations of divergent differentiators and convergent conformists, we identified a third considerable population labelled compromisers. Compromisers tend to provide a balance between free of charge and paid offerings. For example, Spotify allows for music streaming already in its free offering (App Annie, 2015x & Google Play, 2015z). Some more advanced features (e.g. listen to music offline) and an improved user experience without advertisements permit superior levels of convenience that requires premium conversion. Thus, compromisers attempt to moderate and balance the relationship between the free and premium offering, which is why their monetization risk is medium. Compared to conformists, compromisers provide more options to differentiate their premium offering. Still, the free offering takes up a larger portion of the complete offering than the more restrictive differentiator population. Therefore, compromisers are a separate population located between the more extreme conformist and differentiator populations.

The second key theme of the typology is the nature of the premium revenue stream. Since monetization is central to the Freemium business model (e.g. see Wagner et al., 2014) and persistently present in the data sample, we derived premium revenue streams from the dimension revenue models as second distinguishing theme to develop the typology. Table 7 summarizes differentiating criteria of the three revenue stream populations.

Variable stream	**Merged stream**	**Fixed stream**
• Dynamic, unscheduled revenue streams • Irregularly recurring revenue streams if customer is retained • Low predictability of revenue (c.p.) • High upside potential (c.p.)	• Fusion of fixed- and variable-stream populations • Medium upside potential and predictability (c.p.)	• Predetermined, time-based monetization • Constant or linearly recurring revenue stream if customer is retained • High predictability of revenue (c.p.) • Low upside potential (c.p.)

Table 7. Explanations of populations for theme "premium revenue streams"

The fixed stream population subsumes any revenue model which pursues a predetermined, time-based monetization strategy. Either an infinite time horizon or a limited time horizon can be charged. If an infinite time horizon is charged, a non-recurring, one-time payment is made to access premium for an eternal time frame. Hence, the link between time and revenue is constant and does not provide any additional upside potential for the publisher. The app "Slow Motion Video FX", for instance, exclusively offers a one-time charge to remove advertisements (App Annie, 2015al & Google Play,

2015ap). The other fixed-stream option is to charge for a limited time horizon and benefit from a regularly recurring revenue stream if the customer is continuously retained. In this case, the time-based regularity of the payments implies a linear connection between time and revenue. If the customer is satisfied with the premium offering, she will continue paying for it. In addition, there is a lock-in effect for subscriptions (Kim & Yoon, 2004), since the customer is required to pay for the premium offering for a fixed time period and has the convenient option to continue usage and payment thereafter. Payments of the fixed stream population can only be incurred based on a fixed time schedule. Thus, time-related predictability of future revenue is high, while the upside potential is rather low due to the time-based restrictions (c.p.). The revenue models for limited time horizons which we identified in the analyses were flat rate and subscription fees. For example, Freeletics (App Annie, 2015m & Google Play, 2015n) charges a monthly or annual subscription fee for their premium offering, while WhatsApp (App Annie, 2015n & Google Play, 2015o) applies an annually due flat rate. Accordingly, the fixed stream population charges for a time-based consumption of the premium offering with a constant or linearly recurring revenue stream if the customer is continuously retained.

In contrast, the variable stream population subsumes dynamic revenue streams which are unscheduled and whose payment structure tends to be asymmetrically distributed across time. Thus, the variable stream population has irregularly and asymmetrically recurring revenue streams if the customer is retained. Customers continuously decide whether they intend to make any additional purchase. The greater customer freedom significantly reduces the time-related predictability of future revenue to be generated for the publisher (c.p.). At the same time, there is a high upside potential (c.p.) without any time-based restrictions. Hence, customers can make purchases any time they perceive purchasing needs. Variable stream revenue models include any form of dynamically recurring payments like virtual currency, pay-per-use or pay-per-issue. For instance, the game Clash of Clans provides a virtual currency which customers can continuously buy to accelerate the gaming progress (App Annie, 2015ab & Google Play, 2015ad). Overall, customers maintain a high degree of flexibility in their purchases. While publishers have to cope with a low time-related predictability of future revenue, they may also gain from significant upside potential.

Finally, a fusion of fixed and variable stream populations labelled merged stream population can be observed. In this case, users either have a clear choice between fixed and variable revenue streams, or they can convert to premium based on a pre-specified combination of fixed and variable revenue streams. For instance, the dating app Lovoo lets users decide which form of premium revenue stream they prefer, e.g. based on their personal budgets and their individual appreciation of the premium offering (App Annie, 2015t & Google Play, 2015v). Users can choose between variable stream revenue models (here: virtual currency and micro-payments) and a fixed stream model (here: time-dependent flat rate for VIP membership). Moreover, there can be a pre-determined combination of the two original revenue stream populations. For example, the online flea market platform Shpock combines a service-/ time-dependent flat rate with a virtual currency in order to monetize its app (App Annie, 2015v & Google Play, 2015x). Thus, either publishers decide on the combination between fixed and variable stream models or customers decide which revenue stream they favour. Both upside potential and time-related predictability of future revenue are medium compared to fixed and variable stream populations (c.p.).

The six outlined populations form nine distinct Freemium types. Taken together, the populations explain exhaustively each of the nine Freemium types. We designed the typology grid based on the described risk considerations conceptualized in each population (see figure 3). As both high levels of convergence and variable revenue streams are considered more risky than high levels of divergence and fixed revenue streams, risk increases the further one moves to the upper right of the typology grid (viewing the relationship theme horizontally). We labelled the respective types accordingly.

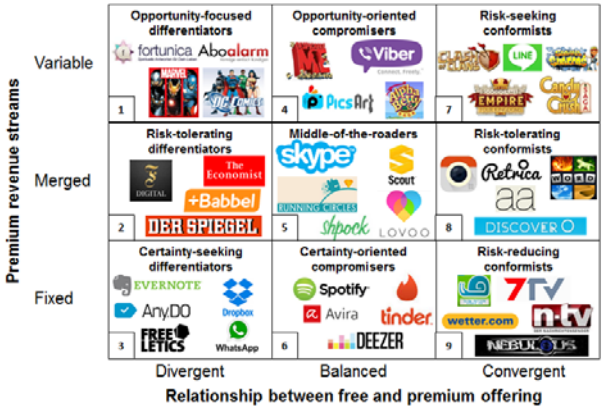

Figure 3. The Typology of Freemium Business Models for Mobile Apps

6 Discussion

6.1 Evaluation of the Freemium typology

The presented typology combines empirical observations with state-of-the-art Freemium literature. Due to the incorporation of premium revenue streams as a distinct theme in the typology, the monetization aspect is made explicit. At the same time, we integrated the Freemium-specific relationship between free and premium offerings. Therefore, this typology captures central elements of the Freemium business model, namely revenue streams and value proposition (see Gassmann et al., 2014; Wagner et al., 2014; Osterwalder & Pigneur, 2010). The selection of these two themes, which are present in any Freemium offering, implies both a parsimonious use of variables and a broad applicability (Glaser & Strauss, 1967) to Freemium offerings which likely goes well beyond mobile apps. By applying the constant comparison method (Glaser & Strauss, 1967), we identified differences and commonalities across Freemium types. Due to its generic nature, the developed Freemium typology is very versatile (Bailey, 1994). The distinction of nine Freemium types enables a significant reduction of complexity (based on Bailey, 1994), while the typology permits an easily understandable overview of Freemium business models. Furthermore, the classification of nine types still ensures manageability (adopted from Bailey, 1994) and provides insights into the diversity of Freemium business models.

Nevertheless, the empirical validity of the typology is restricted to the sampled apps, which limits the general validity of this typology beyond the study context. The respective types need to be filled with further apps to reach a representative sample. Besides, we developed the typology based on observable data, so other important dimensions may not have been considered in the typology building process. The location of individual apps on the theme relationship between free and premium offering requires case-specific judgment. Depending on where the respective analyst draws the line on the continuum, disagreement about which population was identified may arise. Thus, the validity of the typology would also benefit from the independent app assessment of additional coders. Since the typology explicitly focuses on Freemium models and their premium revenue streams, we did not account for additional cash inflows originating from other business models (e.g. e-Commerce for Evernote). In addition, we disregarded non-observable revenue streams due to the chosen research method. Finally, the typology provides a description about Freemium business models for mobile apps (see Bailey, 1994), which means that so far we principally make inferred ceteris paribus statements about the risk inherent in the respective types and their upside potential. The findings at hand cannot provide a theory explaining which type may be adequate given a certain situation.

6.2 Conceptualization

To add a more conceptual perspective to the typology, we formulate three propositions which offer theoretical explanations on the (monetary) advantageousness of certain types within a given

population. To do so, we first clarify the key assumptions that define the circumstances under which the propositions work. First, we assume that the publishers operate in a quasi-monopolistic market (adopted from Cheng & Tang, 2010). Accordingly, they do not have to consider any competitor when making market-related decisions and can focus on the user. Second, we assume that mobile apps can be regarded as experience goods (based on Niculescu & Wu, 2014). Thus, the user benefits from experiencing the free offering before a purchase decision is made (adopted from Nelson, 1970). Third, we assume that publishers and their users generally prefer certainty over risk (adopted from Hirshleifer & Riley, 1992). For publishers, this implies that they prefer fixed over variable revenue streams. Due to the property of apps as experience goods, we assume that users prefer a high convergence of the free and premium offering to be able to experience most of the app offering before making a purchase. Thus, both publishers and users are risk averse (for more derivation of risk aversion, see: Pratt, 1964). Fourth, one of the publisher's main objectives is to maximize sales. Based on these assumptions, we developed three propositions which we present below. These propositions determine the optimal Freemium type given a certain relationship between the free and premium offering. Moreover, we assume that the optimal types, on average, best monetize their offering within each population.

High convergence of the free and premium offering means that the vast majority of the offering is free of charge, so there are few premium restrictions on the app which limit functionality, usage time, user experience, or any combination. Therefore, the main challenge for publishers is to convince users that their premium offering is valuable and desirable to purchase. As most of the offering is free of charge, publishers operate under a high monetization risk (see section 5). Due to the assumed risk aversion, publishers require a risk premium (Pratt, 1964). Since the upside potential for variable revenue streams is the highest across all populations, we assert that variable revenue streams are the optimal choice to match risk and expected return.

> **Proposition 1**: The risk-return profile for the conformist population is best monetized by applying variable revenue streams, so risk-seeking conformists are the optimal type given the high level of convergence.

High divergence of the free and premium offering implies that the majority of the offering requires payment, so there are many restrictions on the app which hinder users in experiencing the app. Therefore, users perceive a high risk before the potential purchase (see e.g. Dowling & Staelin, 1994). The risk of users can be effectively reduced by offering certainty with regard to the revenue stream. As users are commonly risk averse, they prefer scheduled and regular payment structures over unscheduled and unpredictable payments. If users can precisely calculate the total spending, their purchasing risk is reduced. Therefore, fixed revenue streams are the optimal choice in this case.

> **Proposition 2**: For the differentiator population, a fixed revenue stream monetizes best, so certainty-seeking differentiators are the optimal type given the high level of divergence.

A balance between the free and premium offering requires a more complex payment structure in order to account for the diverse needs of user groups. While some users tend to perceive a high purchasing risk, others rather struggle with the differentiation of the premium offering and need to be convinced of its benefits. Since publishers aim at sales maximization, ideally both user groups need to be monetized. Due to the diversity of user group characteristics (e.g. subjective purchasing risk, individual budget constraints), it is most effective to let the users self-select (adopted from Shapiro & Varian, 1998b) their preferred revenue stream.

> **Proposition 3**: The compromiser population best monetizes by applying a merged revenue stream, so middle-of-the-roaders are optimal given their balanced free and premium relationship.

We summarize the optimal types and the respective main challenge in figure 4. These three propositions give clear recommendations on which type is favourable given a certain design of the app offering. The statements constitute one of the first attempts to conceptualize the Freemium business model and provide theoretical explanations for the (monetary) advantageousness of certain types. The focus is on existing apps, as the propositions offer strategic advice on how to design the most adequate revenue stream for a given relationship between free and premium offering. In addition, the

propositions can also be interpreted for new-to-market apps by first deciding on the preferred premium revenue stream and then designing the app offering accordingly. Still, the propositions rely on narrow assumptions, so publishers first need to verify whether the assumptions apply in their specific situation.

Figure 4. Optimal type choices based on propositions

6.3 Future research directions

As shown above, research on Freemium business models is still in an early stage. Therefore, we present some future research areas which go beyond the scope of this paper. Based on an agreement on how to measure monetization, the empirical validity of the theoretically derived propositions can be assessed. The performance of the best-selling apps of each type could be determined in order to make predictions about the advantageousness of certain types. To account for the multi-period distribution of revenue present in most revenue models, it is advisable to perform a customer lifetime value calculation (e.g. see Gupta & Lehmann, 2008). In addition to this across-type performance comparison, a micro-analysis within each type could be performed to assess the specific success factors of each type. For example, it would be interesting to know which revenue model and which limitation mechanism works best for each type. The combination of Freemium and in-app advertisements appeared more regularly than anticipated. Therefore, it may be worthwhile to build upon prior research (e.g. see Riggins, 2003) and study the application of this hybrid model more closely.

7 Conclusion

Based on a sample of 40 apps and a broad literature review, we developed a novel Freemium typology guided by a two-phase research process. By applying theoretical sampling, we observed the most significant dimensions and their characteristics in the exploration phase. As a result of the abstraction phase, we derived the two key themes ‘relationship between free and premium offering’ and ‘premium revenue stream’ as the differentiating variables of the typology. Both academia and our study findings highlight the importance of these two themes, so the Freemium typology is built upon a strong academic and empirical foundation. Since we further distinguished the two themes into six populations that form nine distinct Freemium types, we addressed the main research question. On top of the typology, we formulated three propositions. These propositions give strategic advice on how to position existing Freemium offerings in a quasi-monopolistic market. Moreover, the limitations of both the typology and the formulated propositions were identified and potential research areas were suggested.

To date, the Freemium business model is the most successful direct monetization option for mobile apps. With the on-going rise of the mobile app economy, the importance of Freemium is likely to increase in the future, as well. Therefore, it is important to establish an effective Freemium classification scheme and develop theoretical propositions which may support app publishers in their decision-making. The developed typology contributes to a better understanding of the differentiating criteria of Freemium business models and enables risk-return considerations in the decision-making of app publishers. In addition, the typology at hand effectively reduces the complexity of Freemium business models and provides a generalizable framework for different kinds of Freemium offerings.

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